1 MEASURE

## Auction-grounded compute-utility frontier

216 SCENARIOS · 6 CHECKPOINTS

LEAKAGE-CONTROLLED DECISION PIPELINE

FIXED WITHIN-PRODUCT SPLIT

70% | 30%

**Auction data**

10,681 bids
628 auctions
3 products

Palm · Xbox · Cartier

CONSTRUCTION

Displayed win probabilities

$\widehat{F}_{\text{construction}}(b)$

HELD OUT

Hidden until commitment

Scores the chosen bid

CONTROLLED READOUT

- 3 randomized option orders
- Common option-letter readout
- Scenario-paired bootstrap
- No smoothing or monotonicity rule

**LLM chooses one bid**

private value $v_s$ · bid menu $B_s$

| OPTION A | OPTION ... | OPTION K |
|---|---|---|
| $b_1,\ \widehat{p}(b_1)$ | $K \in \{5,7\}$ | $b_K,\ \widehat{p}(b_K)$ |

EXTERNALLY ASSIGNED REASONING c

0 · 64 · 128 · 256 · 512 · 768

COMMITTED $\widehat{b}$

**Held-out utility score**

Applied only after bid commitment

$$U_s(b \mid v_s) = \widehat{F}_{\text{held-out},s}(b)\,(v_s - b)$$

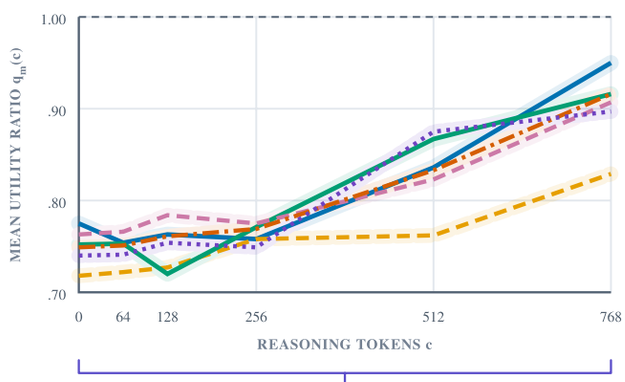
$$Q_{msc} = \frac{U_s(\widehat{b}_{msc} \mid v_s)}{U_s(b_s^* \mid v_s)}$$

MEAN Q OVER 216 SCENARIOS = $q_m(c)$

FROM HELD-OUT SCORES

UTILITY FRONTIERS

6 CHECKPOINTS · 216 SCENARIOS · 3 OPTION ORDERS




ENDPOINT GAIN: $q_m(768) - q_m(0)$

**+0.111 to +0.175**

Range across six checkpoints

CONSISTENT ENDPOINT IMPROVEMENT

Held-out utility ratio improves for all six checkpoints

0 to 768 tokens · all paired 95% CIs > 0

Scenario bootstrap · 2,000 draws

UNCERTAINTY & ROBUSTNESS CHECKS

- 5-seed hierarchical CI
- Auction-CDF bootstrap
- Synthetic IPV exact payoff
- Free-form 5 estimable models

5-SEED STOCHASTIC $q_m(c)$ FOR THE GAME

2 MODEL

## Two-agent compute contest

$C_S = \{128, 384, 768\}$

Modeled agents choose c; evaluated LLMs do not




WIN PROBABILITY

$$P_i(c_i, c_j; \gamma) = \frac{q(c_i)^\gamma}{q(c_i)^\gamma + q(c_j)^\gamma}$$

PRIVATE PAYOFF

$$\pi_i = VP_i - \lambda c_i$$

$$C_{\text{NE}} = 2c_{\text{NE}}$$

SOCIAL PLANNER

$$W_\beta(c_1, c_2) = \beta Q_\gamma - \lambda(c_1 + c_2)$$

$$Q_\gamma = P_1 q(c_1) + P_2 q(c_2)$$

$$C_{\text{OPT}} = c_1^* + c_2^*$$

| $\kappa = \lambda/V$ | $\rho = \beta/V$ | $\gamma$ |
|---|---|---|
| Per-token cost λ relative to prize V | Planner's quality value β relative to V | Winning-odds sensitivity to quality |

3 COMPARE

## Nash vs. planner total compute

| OVERCOMPUTE | ALIGNED | UNDERCOMPUTE |
|---|---|---|
| $C_{\text{NE}} > C_{\text{OPT}}$ | $C_{\text{NE}} = C_{\text{OPT}}$ | $C_{\text{NE}} < C_{\text{OPT}}$ |




Schematic: exact boundaries depend on $q(c)$ and $\gamma$

COMPARATIVE STATICS

- $\gamma$ Private threshold shifts right
- $\rho$ Planner compute rises

COST-AXIS ANCHOR

RTX 4090 rental + mean prize

$\kappa \in [7.50\times10^{-9},\ 1.36\times10^{-8}]$

ALL 6 FRONTIERS: 3 REGIONS · CONDITIONAL; $\rho$ AND $\gamma$ NOT IDENTIFIED